研究室会 発表資料

2026/10/7 Ryo Matsushita

"Don't Think of" vs. "Think of Something Else"

Internal Representations of Thought Suppression and Concept Replacement in LLMs

Suppression

"Don't think of X"

VS

Replacement

"Think of Y instead of X"

How does an LLM internally process a concept it never outputs?

"Don't Think About X" — and You Already Did



"Don't think of a pink elephant."

Don't think of X

↓

Must first recognize X

↓

X gets activated instead?

Thought Suppression / Ironic Process Theory

Wegner et al. (1987) Paradoxical Effects of Thought Suppression

"Don't Think of X" Has Been Studied in LLMs Too

Mann et al. (2025)

Don't Think of the White Bear

- Verified ironic rebound of forbidden concepts
- Analyzed internal suppression/amplification

arXiv: 2511.12381

Zhou et al. (2026)

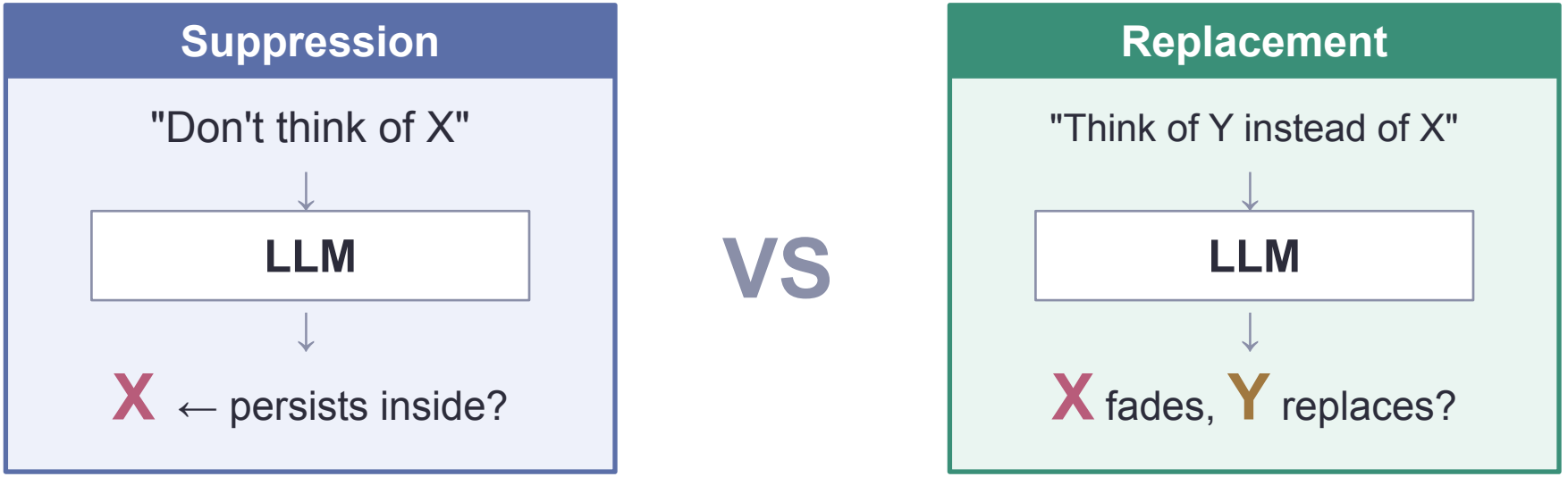
How Language Models Process Negation

- Analyzed how LLMs process negation internally
- Suppression / Constructive Mechanism

ICML 2026 / PMLR

It is now known that forbidden concepts are represented and suppressed internally

Are These Two Instructions Internally Equivalent?



Research Question: How does the internal representation of concept X differ between Suppression and Replacement?

Comparing Across 4 Conditions

